Outils Mathématiques – Calcul Vectoriel

CV1

1) Les forces $\vec{F}_1$ *et* $\vec{F}_2$ ne dépendent d'aucunes variables donc elles sont constantes.

2) $\vec{F} = \vec{F}_1 + \vec{F}_2 = \vec{e}_x + 2\vec{e}_y + 3\vec{e}_z + 4\vec{e}_x - 5\vec{e}_y - 2\vec{e}_z$

$\vec{F} = 5\vec{e}_x - 3\vec{e}_y + \vec{e}_z$

$\|\vec{F}\| = \sqrt{5^2 + (-3^2) + 1^2} = \sqrt{35}$ donc $F = \sqrt{35}\,N$

3) $\overrightarrow{AB} = (0-20)\vec{e}_x + (0-15)\vec{e}_y + (7-0)\vec{e}_y = -20\vec{e}_x - 15\vec{e}_y + 7\vec{e}_z$

$W_{AB} = \vec{F}.\overrightarrow{AB} = 5\times(-20) + (-3)\times(-15) + 7\times 1$

$W_{AB} = -48\,J$

CV3

1) $\|\vec{r}_1\| = \sqrt{4^2 + (-3)^2} = \sqrt{25} = 5$

$\|\vec{r}_2\| = \sqrt{2^2 + (-4)^2 + (-4)^2} = \sqrt{36} = 6$

$\|\vec{r}_3\| = \sqrt{2^2 + (-3)^2 + 6^2} = \sqrt{49} = 7$

2) $\vec{r}_1.\vec{r}_2 = 4\times 2 + (-3)\times(-4) + 0\times(-4) = 8 + 12 = 20$

$\vec{r}_1.\vec{r}_3 = 4\times 2 + (-3)\times(-3) = 8 + 9 = 17$

$\vec{r}_2.\vec{r}_3 = 2\times 2 + (-3)\times(-4) + (-4)\times 6 = 4 + 12 - 24 = -8$

$$\vec{r}_1 \wedge \vec{r}_2 = \begin{vmatrix} 4 & 2 & \vec{e}_x \\ -3 & -4 & \vec{e}_y \\ 0 & -4 & \vec{e}_z \end{vmatrix} = ((-3)\times(-4) - (-4)\times 0)\vec{e}_x + (0\times 2 - (-4)\times 4)\vec{e}_y + (4\times(-4) - 2\times(-3))\vec{e}_z$$

$\vec{r}_1 \wedge \vec{r}_2 = 12\vec{e}_x + 16\vec{e}_y - 10\vec{e}_z$

$$\vec{r}_1 \wedge \vec{r}_3 = \begin{vmatrix} 4 & 2 & \vec{e}_x \\ -3 & -3 & \vec{e}_y \\ 0 & 6 & \vec{e}_z \end{vmatrix} = -18\vec{e}_x - 24\vec{e}_y - 6\vec{e}_z$$

$$\vec{r}_2 \wedge \vec{r}_3 = \begin{vmatrix} 2 & 2 & \vec{e}_x \\ -4 & -3 & \vec{e}_y \\ -4 & 6 & \vec{e}_z \end{vmatrix} = -36\vec{e}_x - 20\vec{e}_y + 2\vec{e}_z$$

3) $\vec{U} = \vec{r}_1 + \vec{r}_2 + \vec{r}_3$: $\vec{U}\begin{pmatrix} 4+2+2 \\ -3-4-3 \\ 0-4+6 \end{pmatrix} = \vec{U}\begin{pmatrix} 8 \\ -10 \\ 2 \end{pmatrix}$ donc $\vec{U} = 8\vec{e}_x - 10\vec{e}_y + 2\vec{e}_z$

$\|\vec{U}\| = \sqrt{8^2 + (-10)^2 + 2^2} = \sqrt{64 + 100 + 4} = \sqrt{168}$

$\vec{V} = \vec{r}_1 + \vec{r}_2 - \vec{r}_3$: $\vec{V}\begin{pmatrix} 4+2-2 \\ -3-4+3 \\ 0-4-6 \end{pmatrix} = \vec{V}\begin{pmatrix} 4 \\ -4 \\ -10 \end{pmatrix}$ donc $\vec{V} = 4\vec{e}_x - 4\vec{e}_y - 10\vec{e}_z$

$\|\vec{V}\| = \sqrt{4^2 + (-4)^2 + (-10)^2} = \sqrt{16 + 16 + 100} = \sqrt{132}$

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4) Méthode courte

$$\vec{U}.\vec{V} = 8\times 4+(-10)\times(-4)+2\times(-10) = 32+40-20 = 52$$
$$\vec{U}\wedge\vec{V} = \begin{vmatrix} 8 & 4 & \vec{e}_x \\ -10 & -4 & \vec{e}_y \\ 2 & -10 & \vec{e}_z \end{vmatrix} = 108\vec{e}_x+88\vec{e}_y+8\vec{e}_z$$

En utilisant les résultats du 2)

$$\begin{aligned}\vec{U}.\vec{V} &= (\vec{r}_1+\vec{r}_2+\vec{r}_3).(\vec{r}_1+\vec{r}_2-\vec{r}_3) = \vec{r}_1{}^2+(\vec{r}_1.\vec{r}_2)+(-\vec{r}_1.\vec{r}_3)+(\vec{r}_2.\vec{r}_1)+\vec{r}_2{}^2+(-\vec{r}_2.\vec{r}_3)+(\vec{r}_3.\vec{r}_1)+(\vec{r}_3.\vec{r}_2)-\vec{r}_3{}^2 \\ &= \vec{r}_1{}^2+\vec{r}_2{}^2-\vec{r}_3{}^2+2(\vec{r}_1.\vec{r}_2) \\ &= 5^2+6^2-7^2+2\times 20 \\ \vec{U}.\vec{V} &= 52\end{aligned}$$

$$\begin{aligned}\vec{U}\wedge\vec{V} &= (\vec{r}_1+\vec{r}_2+\vec{r}_3)\wedge(\vec{r}_1+\vec{r}_2-\vec{r}_3) = \vec{r}_1\wedge\vec{r}_1+\vec{r}_1\wedge\vec{r}_2-\vec{r}_1\wedge\vec{r}_3+\vec{r}_2\wedge\vec{r}_1+\vec{r}_2\wedge\vec{r}_2-\vec{r}_2\wedge\vec{r}_3+\vec{r}_3\wedge\vec{r}_1+\vec{r}_3\wedge\vec{r}_2-\vec{r}_3\wedge\vec{r}_3 \\ &= 0+(\vec{r}_1\wedge\vec{r}_2)-\vec{r}_1\wedge\vec{r}_3+(-\vec{r}_1\wedge\vec{r}_2)+0-\vec{r}_2\wedge\vec{r}_3-\vec{r}_1\wedge\vec{r}_3-\vec{r}_2\wedge\vec{r}_3-0 \\ &= -2(\vec{r}_1\wedge\vec{r}_3)-2(\vec{r}_2\wedge\vec{r}_3) \\ &= -2(-18\vec{e}_x-24\vec{e}_y-6\vec{e}_z)-2(-36\vec{e}_x-20\vec{e}_y+2\vec{e}_z) \\ \vec{U}\wedge\vec{V} &= 108\vec{e}_x+88\vec{e}_y+8\vec{e}_z\end{aligned}$$

CV4

1) $\|\vec{r}_1\| = \sqrt{2^2+3^2+(-1)^2} = \sqrt{14}$ $\|\vec{r}_2\| = \sqrt{3^2+(-2)^2+2^2} = \sqrt{17}$ $\|\vec{r}_3\| = \sqrt{4^2+(-3)^2+3^2} = \sqrt{34}$

2) $\cos(\vec{r}_1,\vec{r}_2) = \dfrac{\vec{r}_1.\vec{r}_2}{\|\vec{r}_1\|\|\vec{r}_2\|} = \dfrac{2\times 3+3\times(-2)+(-1)\times 2}{\sqrt{14}\sqrt{17}} = \dfrac{-2}{\sqrt{238}} \approx -0{,}13$

$\Rightarrow (\vec{r}_1,\vec{r}_2) \approx \pm 97°$

3) $\vec{r}_1\wedge\vec{r}_2 = \begin{vmatrix} 2 & 3 & \vec{e}_x \\ 3 & -2 & \vec{e}_y \\ -1 & 2 & \vec{e}_z \end{vmatrix} = 4\vec{e}_x-7\vec{e}_y-13\vec{e}_z$ *donc* $(\vec{r}_1\wedge\vec{r}_2).\vec{r}_3 = 4\times 4+(-7)\times(-3)+(-13)\times 3 = -2$

$(\vec{r}_1\wedge\vec{r}_2)\wedge\vec{r}_3 \neq 0 \Rightarrow$ les vecteurs ne sont pas coplanaires

4)

CV5

3) $\overrightarrow{CA}(2,-1,-1)$, $\overrightarrow{AB}(-2,-3,2)$

$$\overrightarrow{M_{\overrightarrow{AB}/C}} = \overrightarrow{CA}\wedge\overrightarrow{AB} = \begin{vmatrix} 2 & -2 \\ -1 & -3 \\ -1 & 2 \end{vmatrix} = -5\vec{e}_x-2\vec{e}_y-8\vec{e}_z$$

CV6

1) $\overrightarrow{SA}(a\sqrt{3},-3a,-h)$ $\overrightarrow{SB}(a\sqrt{3},3a,-h)$ $\overrightarrow{SC}(-2a\sqrt{3},0,-h)$

$$\|\overrightarrow{SA}\| = \sqrt{(a\sqrt{3})^2+(-3a)^2+(-h)^2} = \sqrt{12a^2+h^2} = \sqrt{400}$$
$$\|\overrightarrow{SA}\| = 20\ m$$

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$$\|\overrightarrow{SB}\| = \sqrt{(a\sqrt{3})^2+(3a)^2+(-h)^2} = \sqrt{12a^2+h^2}$$
$$= \|\overrightarrow{SA}\|$$
$$\|\overrightarrow{SB}\| = 20\ m$$

$$\|\overrightarrow{SC}\| = \sqrt{(-2a\sqrt{3})^2+(-h)^2} = \sqrt{12a^2+h^2}$$
$$= \|\overrightarrow{SA}\|$$
$$\|\overrightarrow{SC}\| = 20\ m$$

2) $\overrightarrow{SA}.\overrightarrow{SO} = \overrightarrow{SB}.\overrightarrow{SO} = \overrightarrow{SC}.\overrightarrow{SO} = h^2 = 100$

$$\cos(\overrightarrow{SA},\overrightarrow{SO}) = \frac{\overrightarrow{SA}.\overrightarrow{SO}}{\|\overrightarrow{SA}\|\|\overrightarrow{SO}\|} = \frac{100}{20\times 10} = \frac{1}{2}$$

or $\cos(\overrightarrow{SA},\overrightarrow{SO}) = \cos(\overrightarrow{SB},\overrightarrow{SO}) = \cos(\overrightarrow{SC},\overrightarrow{SO}) = \frac{1}{2} \quad \Rightarrow \varphi = \pm 60°$

3) $\vec{F} = \vec{F}_A + \vec{F}_B + \vec{F}_C$

$$= \frac{\overrightarrow{SA}}{2} + \frac{\overrightarrow{SB}}{2} + \frac{\overrightarrow{SC}}{2}$$
$$= \frac{0\vec{e}_x + 0\vec{e}_y + (-3h)\vec{e}_z}{2}$$
$$\vec{F} = \frac{-3h}{2}\vec{e}_z$$

4) $\overrightarrow{M_{\vec{F}_A/0}} = \overrightarrow{OS} \wedge \vec{F}_A$

$$= h\vec{e}_z \wedge \vec{F}_A$$
$$= \begin{pmatrix} \begin{vmatrix} 0 & a\sqrt{3}/2 \\ 0 & -3a/2 \\ h & -h/2 \end{vmatrix} \end{pmatrix}$$
$$= \begin{pmatrix} 3ah/2 \\ ah\sqrt{3}/2 \\ 0 \end{pmatrix}$$

$$\overrightarrow{M_{\vec{F}_B/0}} = \begin{pmatrix} \begin{vmatrix} 0 & a\sqrt{3}/2 \\ 0 & 3a/2 \\ h & -h/2 \end{vmatrix} \end{pmatrix} = \begin{pmatrix} -3ah/2 \\ ah\sqrt{3}/2 \\ 0 \end{pmatrix}$$

$$\overrightarrow{M_{\vec{F}_C/0}} = \begin{pmatrix} \begin{vmatrix} 0 & -a\sqrt{3} \\ 0 & 0 \\ h & -h/2 \end{vmatrix} \end{pmatrix} = \begin{pmatrix} 0 \\ -ah\sqrt{3} \\ 0 \end{pmatrix}$$

$$\overrightarrow{M_{\vec{F}/0}} = \sum_{i=A,B,C} \overrightarrow{M_{\vec{F}_i/0}} = \vec{0}$$

5) $A(ABC) = \frac{\|\overrightarrow{SA} \wedge \overrightarrow{SB}\|}{2}$

$$\overrightarrow{SA} \wedge \overrightarrow{SB} = \begin{vmatrix} a\sqrt{3} & a\sqrt{3} \\ -3a & 3a \\ -h & -h \end{vmatrix} = \begin{pmatrix} 6ah \\ 0 \\ 6a^2\sqrt{3} \end{pmatrix}$$
$$\|\overrightarrow{SA} \wedge \overrightarrow{SB}\| = \sqrt{36a^2h^2+108a^4} = a\sqrt{36h^2+108a^2}$$

$A(ABC) \approx 198{,}5\ m^2$

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